

TABLE 3.1 Your Money Is Split Equally Across Two Investments: What Is the Probability of Earning Less Than a *One-Year* 5% Certificate of Deposit (CD)?

Correlation Between Investments	Probability of Shortfall
100%	36.94%
75%	35.33%
50%	33.43%
25%	31.11%
0%	28.18%
–25%	24.28%
–50%	18.63%
–75%	9.44%
–100%	0.00%

$E[r]=15\%$, $vol=30\%$; probability each asset earns $<5\%$ = 36.94%

Source: Moshe Milevsky and the IFID Centre, 2008.

As you can see, in the most extreme case, when the correlation between the two investments is a perfect 100%, the probability of earning less than the CD is roughly 37%—very high, indeed. If you knew beforehand that there was almost a two-in-five chance that your investment would fare no better than what you could get from a very secure certificate of deposit at your neighborhood bank, you would think twice, wouldn't you?

So it seems that with a perfect, 100% correlation, there are absolutely no benefits to diversification. If you split your money into two baskets that are in exactly the same place (two markets that are perfectly correlated), you have basically kept your eggs in the same basket. By analogy, then, if you diversify into perfectly correlated investments, the odds of earning less than a standard 5% deposit are the same as if you did not diversify.

At the other extreme is the correlation coefficient of -100% . In this case, if you split your money between two investments, the odds of earning less than a 5% CD are reduced to a perfect zero. In other words, you will never do worse than 5%. You have basically taken two risky investments, put them together in your investment portfolio, and created a situation in which you will never lose money. Not bad at all.

Wait a minute, you might be saying. What's the catch? How can I guarantee that you won't lose money?